

Analytical Chemistry (CHEM-319)

Wrong Questions

Q no. 1 Explain the construction and working of Graphite furnace with the help of diagram?

Ans In Atomic absorption spectroscopy, the graphite furnace is an essential component ^{used} for the analyzing trace elements in sample.

* Construction:

1: Graphite tube:-

The heart of graphite furnace, made of high purity graphite. It is designed to withstand high temperature and serves as a container for the sample.

2: Coil or boat Design:-

The graphite tube can be coiled or shaped like a boat, depending on the specific instrumental design.

3: Power supply:-

Provides the necessary electrical energy to heat the graphite tube.

4: Temperature control system:-

It includes temperature

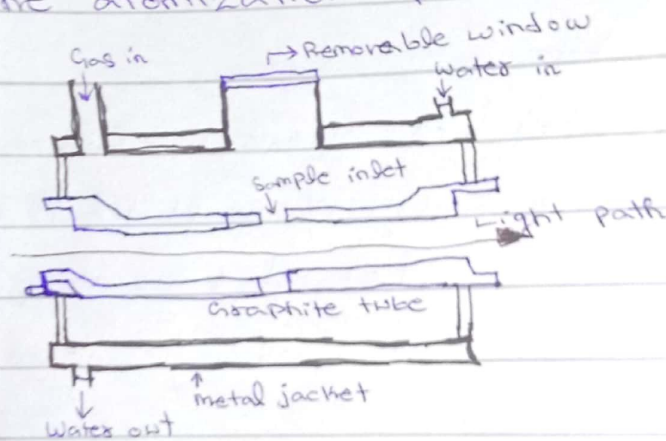
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sensors and a feedback mechanism to precisely control the furnace temperature.

5: Gas flow system:-

Inert gases, usually argon, are introduced to create a controlled atmosphere inside the furnace. These gases protect the sample from oxidation and facilitate the atomization process.



* Working:

1: Drying and Ashing:-

The sample solution, typically containing a dissolved metal or trace element, is introduced into the graphite tube. Initially, the furnace temperature is low, allowing the solvent to evaporate (drying) and the remaining solid components to burn off (ashing).

Atomization:-

The temperature is then rapidly increased to a point where the sample is vaporized and atomized. This atomization step is crucial because it converts the sample into individual atoms in a gas phase. At this high temperature, the atoms are in an excited state.

3: Absorption of Light:-

A hollow cathode lamp emits light specific to the element being analyzed. As the element's atoms are in an excited state, they absorb light at characteristic wavelength when illuminated by this lamp.

4: Detection:-

A detector measures the intensity of light passing through the atomic vapour. The absorption of light at specific wavelengths is proportional to the concentration of the element in the sample.

5. Quantification:-

By comparing the absorption of light intensity to a calibration curve, the concentration of the element in the original sample can be determined accurately.

* The graphite furnace in AAS allows for highly sensitive and selective measurements, particularly useful when dealing with samples containing trace levels of elements. It provides valuable information for various fields, including environmental analysis, clinical research, and material science.



Qno.2 Describe interference and error in AAS?

Ans During the analysis in AAS, several challenges can affect the accuracy and reliability of results.

* Errors in AAS encompass deviations between the measured values and the true values, which can occur due to several

factors, including instrumental limitations, calibration inaccuracies or improper sample preparations. Identifying and minimizing these errors are crucial for obtaining precise and trustworthy analytical results.

* Interference in AAS refers to the phenomenon where substances other than the analyte element influence the measurement process. These interferences can arise from various sources such as the sample matrix, chemical compounds or instrumental factors. Interference can distort the absorption signal, leading to erroneous readings and misinterpretation of the sample's elemental composition.

* Types of interference:-

1. Chemical interference:-

Already written

2. Spectral interference:-

in AAS refers to the

phenomenon where the absorption of light by one element in a sample is affected by the presence of another element, leading to inaccurate concentration measurements. This interference occurs when the absorption lines of different elements overlap in the spectrum, making it difficult to distinguish the absorption signals of the individual elements.

There are several types of spectral interferences:

- * Absorption interference
- * Emission interference
- * Continuum interference

Scientists and researchers use various techniques to minimize or correct spectral interferences in AAS such as using background correction methods, selecting appropriate analytical lines with minimal interference or using chemical modifiers to shift

the absorption lines away from interfering species.

Ionization interference:-

in Atomic absorption occurs when the analyte element or other elements in the sample ionize in the flame, producing ions that can absorb radiation at the same wavelength as the neutral atoms of the elements being analyzed. This phenomenon can lead to inaccuracies in concentration measurements as the absorption signal is affected by the presence of these ions. To minimize ionization interference in AAS, techniques such as matrix modification, wavelength selection, background correction and collision cells are used.

* Matrix effects also cause error in AAS. Differences in the matrix can affect the

atomization and ionization process, causing errors in the measurement.

- * Presence of background absorbance in the flame can interfere with the absorption signal, leading to errors in concentration determination.

- * Problems within the AAS instrument, such as lamp instability, detector issues or misalignment can introduce errors in the analysis.



Q.no.3 Discuss Hydride generation method in detail specially emphasizing on its principle?

Ans Hydride generation is a sample introduction technique used in AAS to enhance the sensitivity and selectivity of the analysis for certain elements.

- * Principle:-

Its principle is based on

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converting certain analyte elements into volatile hydrides, which can be easily transported to and analyzed in the AAS instrument. This technique is primarily used for the element that form volatile hydrides, such as arsenic, antimony and selenium.

* Key steps in the process

1: Sample preparation:-

The sample is typically digested or dissolved in an appropriate acid, depending on the nature of the sample and the element of interest.

2: Hydride formation:-

For elements that can form volatile hydrides (e.g. Arsenic as arsine gas, AsH_3), a reducing agent like $NaBH_4$ is added to the sample solution. This reduces the metal ions to their elemental form and generates hydride gases from hydride forming elements.

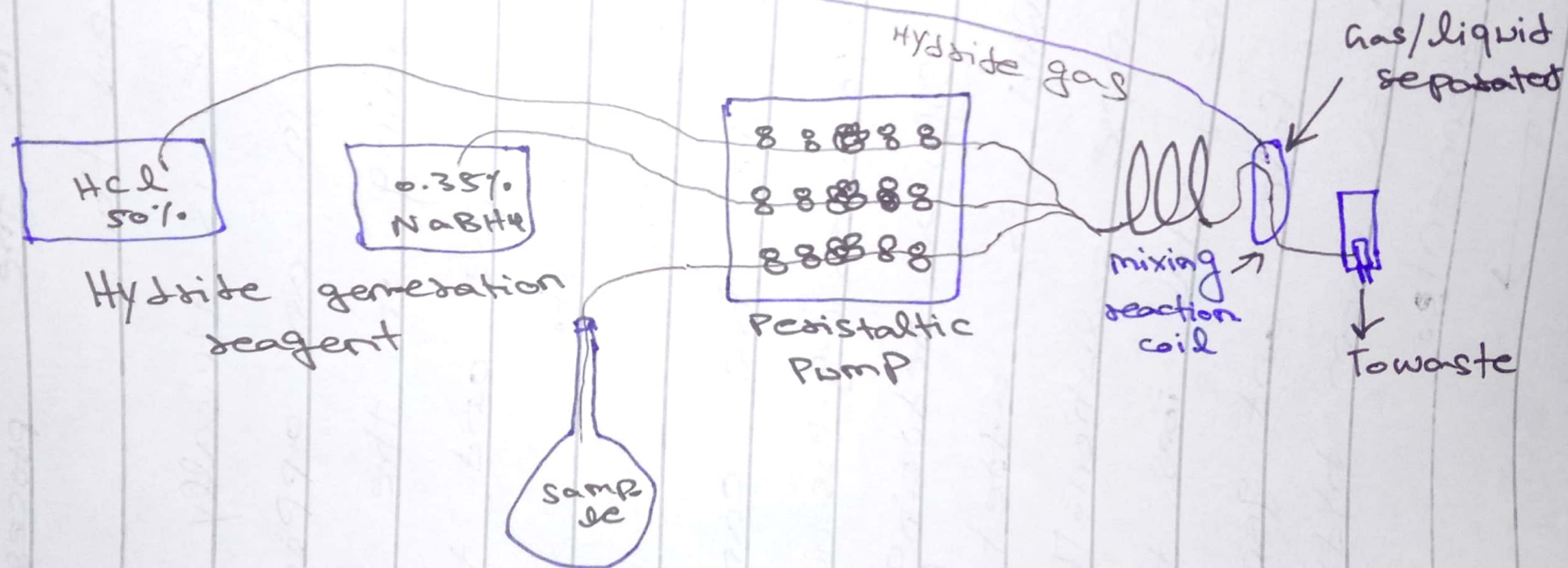
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3: Gas-liquid separation:-

The generated hydride gases are separated from the liquid separated at a gas permeable membrane. This step isolates the volatile hydrides for further analysis.

To optical cell
heated in air/ C_2H_4 flame



Hydride
generator

Transport to AAS Instrument:

The separated hydride gases are then transported to the flame or graphite furnace in the AAS instrument. This is where atomization and excitation of the analyte atoms occur.

5: measurement:

The AAS instrument measures the absorption of the specific wavelength of light by the analyte atoms of hydride gas in the flame or graphite furnace, allowing quantification of the element's concentration in the sample.

* Hydride generation is particularly useful for trace analysis of these elements because it reduces interferences and enhances sensitivity. It is important to note that this

method is specific to certain elements and not applicable to all elements analyzed by AAS.



Umer Abdullah

Govt. Graduate College M.B.Din

Analytical Chemistry (CHEM-319)

Explain zeta potential in capillary zone electrophoresis (CZE)?

Ans In CZE, zeta potential plays a crucial role in understanding the separation and migration of charged molecule within a capillary tube.

Explanation:

*The zeta potential in CZE refers to the electrical potential difference between the bulk solution and the inner surface of the capillary wall. It arises due to the interaction between the charged capillary surface and the surrounding electrolyte solution.

*The zeta potential of the capillary wall affects the magnitude and direction of the electro-osmotic flow. If the zeta potential value of capillary wall is negative, it will attract positively charged ions from the solution, resulting in a net flow of solution towards cathode. Conversely, if capillary wall has a positive zeta potential, it will attract negative

ly charged ions, causing the solution to move toward the anode

*Controlling the zeta potential of the capillary wall is important in CZE because it can influence the separation and resolution of analyte molecules. By adjusting the pH and ionic strength of the electrolyte solution as well as modifying the capillary surface, analysts can manipulate the zeta potential to optimize separation conditions.

*In summary, zeta potential in CZE is related to the electrical potential at the capillary wall, which influences the electroosmotic flow, and consequently, the separation and migration of the charged molecules within the capillary tube. Proper control and understanding of zeta potential are essential for achieving efficient and effective separation in CZE.

